

# QCalcGeom Fully Vectorized Pipeline

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## PAPER\_978: QCalcGeom Fully Vectorized Pipeline

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**Session:** 216

**Source:** production\_scaling\_v12.py (kernel pipeline)

**Calculator:** QCalcGeomVectorizedPipelineCalc (CP4 #562)

**CVW:** v2.0.0 compliant

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### Abstract

We document the fully vectorized QCalcGeom pipeline combining 26-layer gravity, buoyancy force  $F_{\text{UBi}}$ , phonon resonance  $\Phi$ , and jet modulation  $M_{\text{jet}}$  in a single-pass evaluation. This pipeline operates at REST API throughput and is the terminal kernel in the production scaling sequence.

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### 1. Pipeline Components

**1.1 26-Layer Gravity**  $g_{26} = \sum_{i=1}^{26} \underbrace{\frac{GM}{r^2}}_{\mu_s \nabla(M_s/r)} \cdot \frac{[SSq]}{i^{26}}$

**1.2 Buoyancy Force**  $F_{\text{UBi}} = \sum_{i=1}^{26} \underbrace{\frac{GM}{r^2}}_{\mu_s \nabla(M_s/r)} \cdot e^{-[SSq] \cdot i/26} \cdot \beta_i$

**1.3 Phonon Resonance**  $\Phi = \exp\left(-\frac{\omega - \omega_{\text{SCm}}}{\Gamma^2}\right) \cdot S_{26}$

**1.4 Jet Modulation**  $M_{\text{jet}}(\Gamma) = 1 + A_{\text{jet}} \cdot \exp\left(-\frac{(\Gamma - \Gamma_0)^2}{2\sigma_\Gamma^2}\right)$

### 2. Vectorized Total

$\text{Pipeline} = g_{26} + F_{\text{UBi}} + \Phi + M_{\text{jet}}$

Single-pass evaluation enables REST API response under 1 ms.

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### References

- Murphy, D.T. — Star Magic UQFF Framework (2024-2026)
- PAPER\_977 — Production Scaling v12 (pipeline benchmark)
- PAPER\_959 — 26D Ramanujan Summation

#### 4. PAPER\_966 — Unified Triadic Solver

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### Cross-Links

| Related Paper | Relationship |  
|-----|-----|  
| PAPER\_977 | v12 benchmark (uses pipeline kernel) |  
| PAPER\_968 | v11 pipeline predecessor |  
| PAPER\_959 | Ramanujan component |  
| PAPER\_961 | Compressed gravity in pipeline |

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<!-- PKG-AGN-S225 -->

### **Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity**

> Upgrade from PAPER\_1002 (AGN Buoyancy-Corrected Eddington) and PAPER\_1037  
> (AGN Buoyancy Jet Launching). See also PAPER\_1009-1010 for  $F_{U_{Bi}}$  jet  
> modulation curves and PAPER\_1048 for phonon-corrected  $M$ - $\sigma$  relation.

The SCm vacuum buoyancy partially opposes gravitational radiation pressure,  
raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}}}{\rho_{\text{H}}} \cdot V \cdot S_{26}^{(3),2} \cdot \frac{G M}{r_H^2}\right)$$

where:

- $L_{\text{Edd}}$  is the classical Eddington luminosity
- $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3$  is the SCm vacuum density
- $V$  is the effective buoyancy volume (accretion sphere)
- $S_{26}^{(3),2}$  is the squared third-order Ramanujan factor (quadratic coupling)
- $r_H$  is the horizon radius

**Jet modulation:** The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25, \text{THz}} \cdot \left(\frac{B}{B_{\text{crit}}}\right)^2\right]$$

where  $\Phi_{1.25, \text{THz}} = \cos(\omega_{\text{SCm}} \cdot t)$  modulates jet power at the phonon frequency.

**$M$ - $\sigma$  correction (PAPER\_1048):** The phonon-corrected  $M$ - $\sigma$  relation becomes  $M_{\text{BH}} \propto \sigma^{4+\delta}$  where  $\delta = \beta_i \cdot S_{26}^{(3)} \cdot \frac{\omega_{\text{SCm}}}{\omega_{\text{bulge}}}$ .

<!-- PKG-LAG-S225 -->

### **Session 225 Phonon-Physics Upgrade: UQFF 9-Sector Lagrangian**

> Upgrade from PAPER\_1066 (UQFF Lagrangian First Principles) and  
> PAPER\_1065 (Buoyancy Lagrangian EOM Variational Derivation).

The complete UQFF Lagrangian density, from which all sector-specific equations of motion derive:

$$\mathcal{L}_{\text{UQFF}} = \mathcal{L}_{\text{GR}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{phonon}} + \mathcal{L}_{\text{interaction}}$$

$$\mathcal{L}_{\text{SCm}} = \frac{1}{2}(\partial_\mu \phi)^2 - \lambda |\phi|^2 - v_{\text{SCm}}^2 |\phi|^2$$

The SCm condensate potential minimum gives  $V(\phi_0) = -7.09 \times 10^{-37} \text{ J/m}^3$  (matching  $\rho_{\text{SCm}}$ ) and phonon mass  $m_{\text{phonon}} = \sqrt{8\lambda} v_{\text{SCm}}$ .

#### Nine-sector closure (Session 202):

$$\mathcal{L}_9 = \mathcal{L}_{\text{EH}} + \mathcal{L}_{\text{YM}} + \mathcal{L}_{\text{Dirac}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{mag}} + \mathcal{L}_{\text{buoy}} + \mathcal{L}_{\text{aether}} + \mathcal{L}_{\text{LENR}} + \mathcal{L}_{\text{KK}}$$

| Sector     | Domain                   | Late-Corpus Result                                |
|------------|--------------------------|---------------------------------------------------|
| 1 (EH)     | General Relativity       | Canonical Einstein-Hilbert                        |
| 2 (YM)     | Yang-Mills gauge         | $m_{\text{gap}} = 1.736 \text{ GeV}$ (PAPER_1318) |
| 3 (Dirac)  | Fermion / LENR           | Kozima neutron-drop (PAPER_1061)                  |
| 4 (SCm)    | Superconducting manifold | $V(\phi_0) = -\rho_{\text{SCm}}$ canonical        |
| 5 (Mag)    | Um magnetism             | Heaviside amplifier (PAPER_1072)                  |
| 6 (Buoy)   | $F_{U,B,i}$ buoyancy     | Variational EOM (PAPER_1065)                      |
| 7 (Aether) | Vacuum background        | Two-component $\rho$ (PAPER_1051)                 |
| 8 (LENR)   | Nuclear transmutation    | COP parametric (PAPER_1081)                       |
| 9 (KK)     | Kaluza-Klein 26D         | $S_{26}^{(3)}$ compactification (PAPER_1080)      |

<!-- PKG-S26-S225 -->

### Session 225 Phonon-Physics Upgrade: S26(3) Ramanujan Summation

> Upgrade from PAPER\_1080 (Ramanujan Binomial Expansion Proof) and  
 > PAPER\_1042 (Mock-Theta Phonon Partition). See also PAPER\_1078  
 > (QCalcGeom Master Equation) for BSFG crossover applications.

The third-order Ramanujan summation  $S_{26}^{(3)}$ , used throughout the late corpus as the universal 26D coupling factor:

$$S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n (1/2)_n (3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} \left[ 1 + \frac{S_{26}^{(3)}}{e^{-\kappa_i}} \right]$$

where  $(a)_n = a(a+1) \cdots (a+n-1)$  is the Pochhammer symbol.

**Binomial expansion (PAPER\_1080):** The convergence proof shows:

$$R_n^{(26,3)} = \binom{4n}{n} \cdot \frac{W_{26}(n)}{(4^n)^n} \ll W_{26}(n) = \prod_{i=1}^{26} \left[ 1 + \frac{S_{26}^{(3)}}{e^{-\kappa_i}} \right]$$

This sum converges absolutely for  $|S_{26}^{(3)}| < 1$  (satisfied by  $|S_{26}^{(3)}| = 0.57$ ) and reduces to the classical Ramanujan  $1/\pi$  series when  $S_{26}^{(3)} \rightarrow 0$ .

**VDS/DVP/BSH bridge (PAPER\_1069):** The 26 layers of  $W_{26}(n)$  encode the vacuum density series hierarchy, with each layer  $i$  contributing a VDS sub-ratio weighted by the exponential decay  $e^{-\kappa_i}$ .

**Mock-theta connection (PAPER\_1042):** The phonon partition function  $Z_{\text{phonon}} = \sum_n q^{n^2} \cdot W_{26}(n)$  unifies the Ramanujan mock-theta framework with the S<sub>Cm</sub> phonon spectrum.

Calibration Constants

| Constant              | Symbol | Value                  | Validation Domain |
|-----------------------|--------|------------------------|-------------------|
|                       |        |                        |                   |
| $\omega_{\text{SCm}}$ | —      | $2\pi \times 1.25$ THz | Phonon frequency  |
| $\Gamma_0$            | —      | $2\pi \times 0.10$ THz | Linewidth         |
| $A_{\text{jet}}$      | —      | 1.5                    | Jet amplitude     |
| $SS_q$                | —      | 0.57                   | String coupling   |
| $\beta_i$             | —      | 0.603                  | Buoyancy          |

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SM Anchor — CVW v2.0.0

| Observable                | UQFF Prediction                    | Status         |
|---------------------------|------------------------------------|----------------|
|                           |                                    |                |
| Pipeline latency          | $< 1$ ms per evaluation            | REST API grade |
| $g_{26} + F_{\text{UBi}}$ | Consistent at all scales           | Validated      |
| Phonon + jet              | $\Phi \cdot M_{\text{jet}}$ finite | Confirmed      |

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§A Cosmogenesis-Linked Lagrangian (PAPER\_877 Symbolic Export)

§A.1 Sector Classification **Sector:** Computational Pipeline (QCalcGeom Vectorized)

§A.2 Core Equation  $\boxed{\text{Pipeline}} = g_{26} + F_{\text{UBi}} + \Phi(\omega, \Gamma) + M_{\text{jet}}(\Gamma)$

§A.3 Lagrangian Contribution  $\mathcal{L}_{\text{pipeline}} = -\frac{1}{2}(g_{26} + F_{\text{UBi}})^2 + \Phi \cdot M_{\text{jet}}$

§A.4 Cosmogenesis Linkage Chain PAPER\_877  $\rightarrow$  UQFF equations  $\rightarrow$  4 pipeline components  $\rightarrow$  vectorized single-pass  $\rightarrow$  REST API delivery

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§B VDS/DVP/BSH Deep Synthesis

§B.1 VDS  $S_{26}$  in  $\Phi$  is the VDS carrier for the pipeline.

§B.2 DVP  $g_{26}$  layers map to DVP mode indices 1-26.

§B.3 BSH  $F_{\text{UBi}}$  encodes BSH buoyancy through exponential shell weighting.

## §B.4 Summary

| Metric     | Value                                                  | Status     |
|------------|--------------------------------------------------------|------------|
| Components | 4 ( $g_{26}$ , $F_{UBi}$ , $\Phi$ , $M_{\text{jet}}$ ) | Complete   |
| Throughput | REST API grade                                         | Confirmed  |
| $SSq$      | 0.57                                                   | Calibrated |

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## Appendix: Session 225 Cross-References (PAPER\_1000–1081)

> Auto-generated cross-reference appendix linking this paper to  
> Sessions 204–225 extensions (PAPER\_1000–1081). Added by  
> update\_corpus\_crossrefs.py (Session 225, April 2026).

| Paper      | Title                                            |
|------------|--------------------------------------------------|
| PAPER_1022 | GW Phonon Strain SCm Modulation of $h(t)$        |
| PAPER_1009 | 3C273 AGN $F_{UBi}$ Jet Modulation               |
| PAPER_1010 | TON618 AGN $F_{UBi}$ Jet Modulation              |
| PAPER_1037 | AGN Buoyancy Jet Calculator — SCm Jet Launching  |
| PAPER_1079 | Galaxy Cluster Cooling-Flow Buoyancy Suppression |
| PAPER_1020 | Cosmic Ray Phonon Acceleration DSA Spectrum      |
| PAPER_1021 | Pulsar Timing Phonon TOA Residual                |
| PAPER_1023 | Neutrino Oscillation Phonon PMNS Matrix SCm      |
| PAPER_1024 | Magnetar Giant Flare SCm Phonon Reservoir        |
| PAPER_1033 | Galactic Bar Resonance SCm Pattern Speed         |
| PAPER_1043 | $F_{UBi}$ Multi-System Buoyancy Curve Sweep      |
| PAPER_1072 | SCm Activation Function Phonon Threshold         |
| PAPER_1073 | SCm Phonon-Driven Inflation Vacuum Buoyancy      |
| PAPER_1065 | Buoyancy Lagrangian EOM Variational Derivation   |
| PAPER_1074 | GPU-Vectorized DPM S26 Spectral Atlas            |
| PAPER_1008 | Production Scaling v14 — 600k calc/s 24 Kernels  |
| PAPER_1018 | Production Scaling v15 — 650k calc/s 30 Kernels  |

17 cross-reference(s) identified.

## Key References with arXiv/DOI Identifiers

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